

Customer Segmentation & CLV Prediction

RFM Clustering | Lifetime-Value Modeling | Cohort Retention | Sales Trends

Executive Summary

This project analyses **1.06M+ real transactions** from a UK online retailer to answer two questions: *who are our customers today* and *what will they be worth tomorrow*. Using RFM analysis with K-Means clustering, customers were grouped into four actionable segments. Probabilistic BG/NBD and Gamma-Gamma models then predicted each customer's future lifetime value, while cohort and sales-trend analysis revealed retention and seasonal patterns.

Dataset & Approach

Source: Online Retail II (UCI ML Repository) — a real UK online retailer, Dec 2009 to Dec 2011. After cleaning (removing missing customer IDs, cancelled orders and returns), the dataset reduced from **1,048,575** to **793,309** clean transactions covering **5,860 unique customers**.

Customer Segments (K-Means, K = 4)

Segment	Customers	Profile
Champions	1,178	Recent, frequent, high spend (~£10.8K avg)
New / Promising	1,267	Recent buyers, still low frequency
At-Risk	1,452	Were valuable, now slipping away
Lost / Hibernating	1,963	One-time buyers, long inactive

Headline insight: Champions are the smallest group (1,178) yet generate ~73% of total revenue — losing one Champion is like losing ~33 Lost customers.

Customer Lifetime Value (CLV) Prediction

Two probabilistic models were combined to forecast each customer's value over the next 3 months: the **BG/NBD model** predicts *how many* future purchases a customer will make, and the **Gamma-Gamma model** predicts the *average value* of those purchases. Together: **CLV = BG/NBD × Gamma-Gamma**. This lets marketing budget be allocated toward the customers most likely to deliver future revenue.

Cohort Retention & Sales Trends

Cohort analysis (grouping customers by their first-purchase month) showed that retention drops sharply after the first month — most first-time buyers do not return immediately — while a loyal core continues purchasing for many months. **Sales trends** are strongly seasonal, building toward the Q4 holiday period, and a small set of decorative / gift products drives a large share of total units sold.

Marketing Recommendations

Segment	Recommended Action
Champions	Reward & protect — VIP perks, loyalty rewards, referral requests

New / Promising	Nurture toward a 2nd / 3rd purchase with welcome offers
At-Risk	Urgent win-back — personalised 'we miss you' discounts
Lost / Hibernating	Low-cost reactivation only; do not over-invest

Tech stack: Python · Pandas · scikit-learn (K-Means) · lifetimes (BG/NBD, Gamma-Gamma) · Matplotlib · Seaborn

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